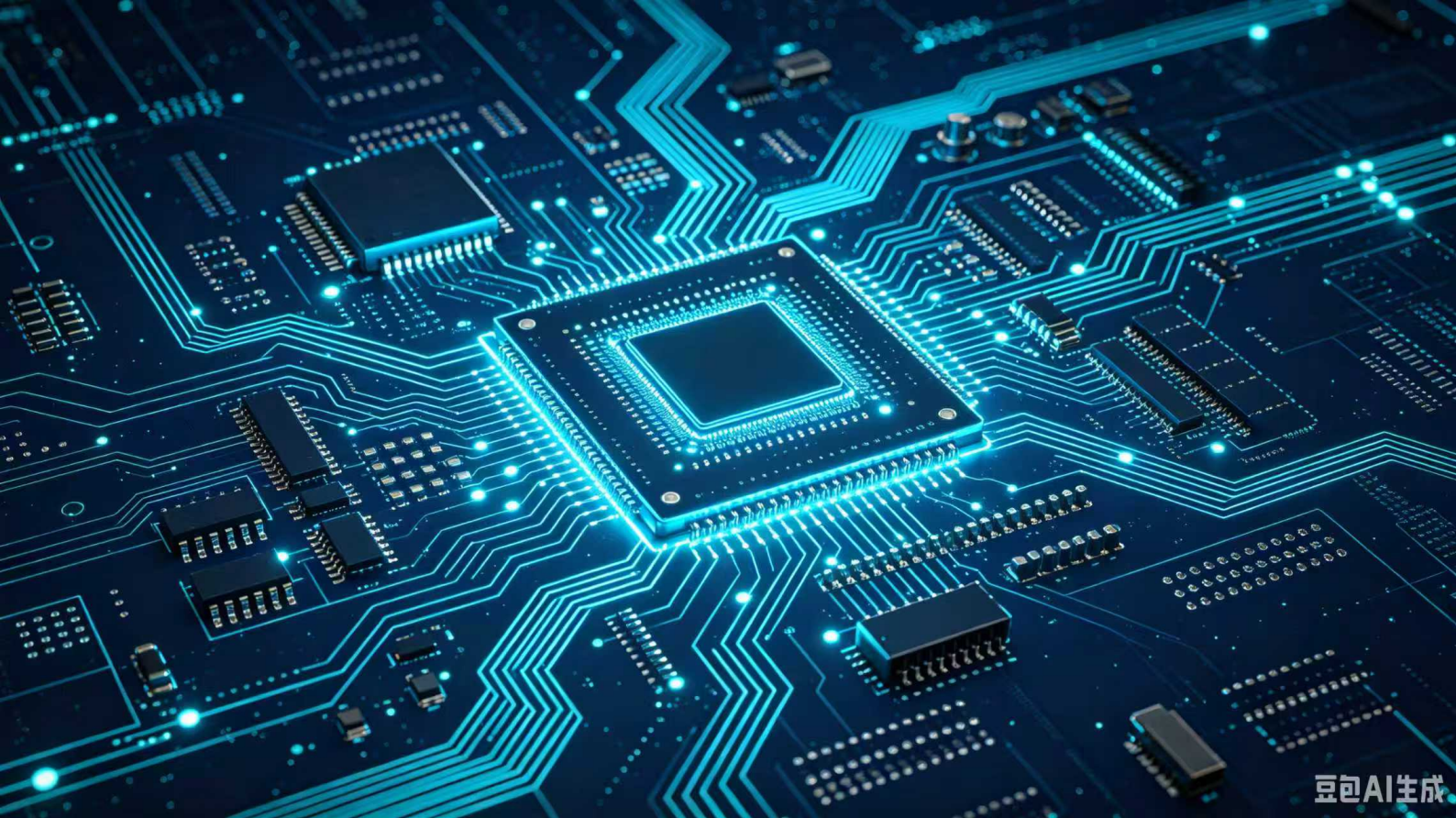




5. Conductors in Electrostatic Field

Nothing is too wonderful to be true, if it be consistent with the laws of nature.

– *Michael Faraday*

All materials are made of atoms. When the atoms come together to form a solid, it is possible that some of the electrons become free to wander about within the solid, while the nucleus and other electrons form the backbone structure of the solid. In such cases, the free electrons serve as “charge carriers”, and an electric current flows when the material is connected to both ends of a battery. Such materials are then called “conductors”. You are familiar with copper, iron and perhaps graphite as examples. In some other cases, there are no “free” electrons inside the solid, and thus no electric current can go through the solid when it is connected to a battery. These are then called “insulators”, and familiar examples are glass, plastic, rubber, and diamond. You may have already noticed that both graphite and diamond are actually formed by carbon atoms only, but their behaviors are sharply different from each other. Thus, not only the atoms forming the materials, but also the way they are brought together, are important to determine whether they belong to insulators or conductors. To fully understand the transport behavior, one needs quantum mechanics and solid-state physics knowledge, which we shall not discuss here.

5.1 Electrostatic Properties of Conductors

The first chapter showed that all materials are made of atoms, each consisting of a nucleus surrounded by electrons. In a conductor some of those electrons are free to travel through the structure, so they can form an electric current under an external electric field. At this stage, we shall not examine the dynamic process with explicit electric current. Instead, we shall look at an isolated conductor in an electrostatic field, such as the one shown in Fig. 5.1.

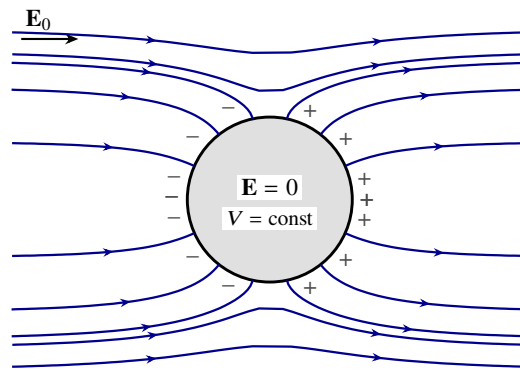


Figure 5.1: An isolated conductor in an electrostatic field. The field inside must be 0 since the conductor itself is an equipotential; the surface charges rearrange until the field just outside meets the surface at right angles.

Imagine we put an isolated conductor inside an electrostatic field. The free electrons immediately “feel” the existence of the field and start moving, and the charges inside the conductor begin to redistribute. In more detail, the free electrons move toward the higher potential, leaving a deficit of electrons, that is, a net positive charge, where they came from. However, since the conductor is isolated, there cannot be a steady current. Instead, after sufficiently long time, the current disappears and the charge distribution also becomes stable. We call the conductor in “equilibrium state” once its charge distribution becomes stable and does not vary in time (hence there is no current). In reality, the transient procedure before the equilibrium state is very fast. We now consider an isolated conductor in an electrostatic field in equilibrium. Firstly, the electric field inside the isolated conductor must be 0, thus the whole conductor is an equipotential. Otherwise, the force due to such a field on free electrons will lead to current inside the conductor, which contradicts the “equilibrium state” assumption. Secondly, by applying Gauss’ law, Eq. (2.1), on any Gaussian surface inside the conductor, we immediately know that there is no “net charge” (charge neutral everywhere) inside the conductor, and all net charge must lie on the surface of the conductor. In the above statements, “inside” means completely surrounded by the metallic material forming the conductor.

For any conductor in electrostatic equilibrium, these are the principles to follow. These two principles are the starting point for the examples below:

Example 5.1: Conductor with cavity.

Assuming we have an isolated conductor with a cavity inside, a similar discussion immediately tells us that the $\mathbf{E}$ -field is 0 everywhere inside the conductor. Therefore, everywhere inside the conductor must be charge neutral, and net charge can only stay on either the inner surface or the outer surface. We now imagine a Gaussian surface around the cavity, making sure that every point on the surface is entirely inside the conductor; see Fig. 5.2. Since the $\mathbf{E}$ -field is zero everywhere on the surface, the total electric flux is also zero for any surface around the cavity inside the conductor. This is only possible if there is no net charge on the inner surface. All the net charge must therefore be on the outer surface.

This alone does not show that there is no charge anywhere on the inner surface. The empty cavity contains no charge, and its wall, being part of the conductor, is at the conductor's potential V_0 . The value of V_0 need not be known: whatever it is, the constant $V = V_0$ satisfies Poisson's equation in the cavity and equals V_0 on its wall, so by the uniqueness theorem stated in Sec. 5.2 it is the potential in the cavity. Hence $\mathbf{E} = 0$ everywhere in the cavity, and a small Gaussian surface straddling the wall shows that $\sigma = 0$ at every point of it.

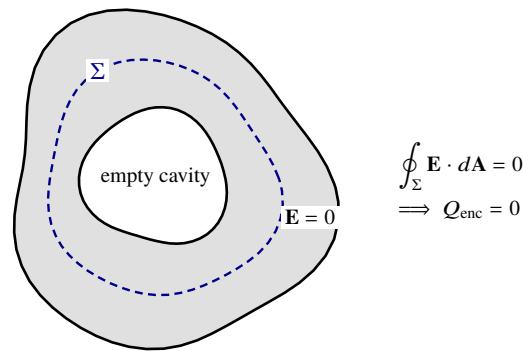


Figure 5.2: An isolated conductor with an empty cavity. The dashed Gaussian surface lies entirely within the conductor where $\mathbf{E} = 0$, so no net charge resides on the inner cavity wall.

The above argument suggests one can screen external electric field using metallic containers. Note that the cavity may contain conductors of its own, as in Fig. 5.3: the wall still carries the induced charge, and whatever happens inside, the field outside depends only on the total charge. What if we want to screen the electric field produced by some charge distribution?

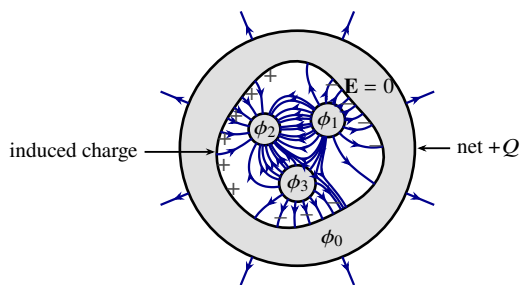


Figure 5.3: Several charged conductors, held at potentials ϕ_1 , ϕ_2 and ϕ_3 , sit inside the cavity of an isolated conductor. The cavity wall carries the induced charge, and the outer surface, at ϕ_0 , responds only to the total charge enclosed.

Example 5.2: Conductor with cavity and additional charge inside cavity.

We now assume that we have an isolated conductor with a cavity inside, but now we have some additional charge inside the cavity. First of all, it is still true that the E -field must be 0 everywhere inside the conductor. Again, we must point out that “inside” means completely surrounded by the metallic material forming the conductor, which does not include the space inside the cavity. Now, if we take a Gaussian surface around the cavity, we again find that the total electric flux through the surface must be 0, if the surface itself is also entirely inside the conductor. That means, if the total net charge inside the cavity $Q \neq 0$, there must be a total charge of $-Q$ on the inner surface of the cavity. The conductor’s total charge does not change, so if the conductor was uncharged its outer surface carries $+Q$; if it carries a net charge Q_c , its outer surface carries $Q_c + Q$. Figure 5.4 shows the uncharged case.

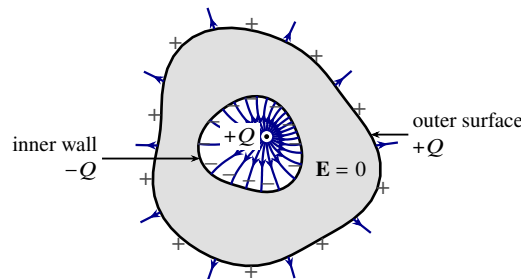


Figure 5.4: An isolated, uncharged conductor with a cavity containing a charge $+Q$. The inner cavity wall acquires $-Q$, and the outer surface acquires $+Q$.

We now imagine taking a conductor with a cavity, and *grounding* it. By *grounding*, we mean connecting the conductor to the earth, which holds it at the earth’s potential, taken as zero — the same as the potential at infinity, since the earth is so large that its potential changes negligibly when charge flows to or from it. If there are no other charges outside, the region outside the container contains no charge and has $V = 0$ on its boundary and at infinity, so $V = 0$ and $\mathbf{E} = 0$ there, and the outer surface carries no charge. If there are charges outside, the outer surface carries the charge they induce, but the field outside is still the same whatever charges are inside. Either way, the electric field of a specific charge distribution can be screened by putting a grounded conducting container around the charge distribution.

You may ask, how are the charges distributed on the surface of the conductor? Two further properties follow. Firstly, the electric field at the surface of an isolated conductor must be perpendicular to that surface; otherwise the parallel component would exert a force on the surface charge and drive a surface current. Secondly, applying Gauss’ law, Eq. (2.1), to an infinitesimally small pillbox straddling the surface gives the magnitude of the field just outside the conductor in terms of the surface charge density there, $E_{\perp} = \sigma / \epsilon_0$.

For the simplest case of an isolated spherical conductor with no other charges nearby, the charge distributes uniformly over the spherical surface by symmetry. For a general shape, the charge distribution is difficult to derive directly. Applying Laplace’s equation together with the principles above, one can solve the problem numerically. Generally, the magnitude of the surface charge density, and with it the strength of the field just outside, is large near sharp outward points and edges of a conductor and small near its inward corners.

5.2 Method of Image Charges

In the most general case, one must solve Laplace's equation numerically to find the charge redistribution and the electric field for an equilibrium conductor in an electrostatic field. In the region outside the conductor there is no volume charge — all the net charge sits on the surface — so the potential there satisfies $\nabla^2 V = 0$, the $\rho = 0$ case of Eq. (4.3). Two features of this equation make the solution guessable, and neither of them is its order. The first is **uniqueness**: in a region whose charge is given, a potential that meets the boundary conditions — the given potential on each bounding surface, or the equipotential condition on the surface of a conductor whose total charge is given instead, together with $V \rightarrow 0$ at infinity if the region is unbounded — is *the* potential there, so any configuration that reproduces those conditions reproduces the field. The second is **linearity**: solutions may be superposed, so images can be built up one boundary at a time. Together they turn a lucky guess into a proof, and they are the idea behind the method of “image charges”. We shall illustrate this method in several examples.

Example 5.3: a point charge q located d above an infinite grounded conducting plane.

In order to solve this problem, we shall assume that the plane is located at $z = 0$, and the point charge is at $(0, 0, d)$. By definition, the infinitely large grounded conducting plane at $z = 0$ ensures that $V(z = 0) = 0$. Therefore, in the lower half-space ($z < 0$), which is filled by the grounded conductor, $\mathbf{E} = 0$: the conductor is an equipotential at $V = 0$, the same potential as infinity, so there is no field anywhere in that region. In the upper half-space ($z > 0$) we introduce an auxiliary system S' with two point charges, namely $+q$ at $(0, 0, d)$ and $-q$ at $(0, 0, -d)$. In S' the charge distribution in the upper half-space ($z > 0$) is exactly the same as in the original system, and $V(z = 0) = 0$ is satisfied there as well. By the uniqueness argument above, the electric field in the upper half-space ($z > 0$) must therefore be the same. That means:

$$V(x, y, z > 0) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} + \frac{-q}{\sqrt{x^2 + y^2 + (z + d)^2}} \right]$$

The electric field can therefore be obtained by $\mathbf{E}(x, y, z > 0) = -\nabla V$.

As a special case, we take a look at $\mathbf{E}(x, y, z = 0^+)$, where 0^+ means approaches 0 from above. The opposite side $\mathbf{E}(x, y, z = 0^-) = 0$ is obvious. Close to the $z = 0$ plane, the $\mathbf{E}$ field is always perpendicular to the xy -plane, because the plane is an equipotential; thus only the $\hat{z}$ component is important. Hence

$$\mathbf{E}(x, y, z = 0^+) = -\hat{z} \frac{\partial V}{\partial z} = -\hat{z} \frac{1}{2\pi\epsilon_0} \frac{qd}{(x^2 + y^2 + d^2)^{3/2}}$$

This quantity is useful for evaluating the **induced charge** at the $z = 0$ surface. Let's imagine an infinitesimal cylindrical Gaussian surface, whose two end surfaces are parallel to the surface we are interested in, and let $\hat{n}$ be the unit normal pointing out of the conductor, that is, $\hat{n} = \hat{z}$. Then

$$\hat{n} \cdot (\mathbf{E}(z = 0^+) - \mathbf{E}(z = 0^-)) = \frac{\sigma}{\epsilon_0}$$

Since $\mathbf{E}(z = 0^-) = 0$, this gives $\sigma = \epsilon_0 E_z(x, y, 0^+)$, so the induced surface charge density is

$$\sigma(x, y) = -\frac{qd}{2\pi(x^2 + y^2 + d^2)^{3/2}}$$

It is negative for $q > 0$, and its total over the plane is $-q$, the charge of the image. Figure 5.5 shows the field lines of this configuration.

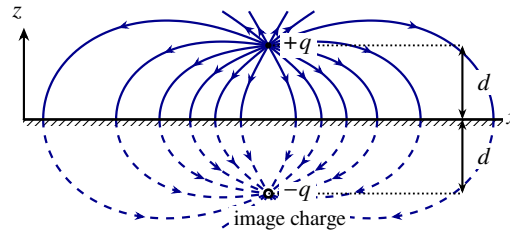


Figure 5.5: The image-charge method. A point charge $+q$ sits a distance d above an infinite grounded conducting plane at $z = 0$. The solid curves are the true field lines in the upper half-space: they leave $+q$ and meet the plane at right angles, because the plane is an equipotential. The dashed curves are their continuations into the region below the plane, where they converge on the image charge $-q$ at $z = -d$. Replacing the plane by that image reproduces $V = 0$ on $z = 0$, and by uniqueness the field above the plane is unchanged.

In the above solution, we see that in order for the method of image charges to work, there are several rules:

1. The image charge cannot appear in the space of interest, that is, the region where we are solving for the field. In the above example, when we were dealing with the upper half-space, the image charge is located in the lower half-space, and the upper half is left unchanged.
2. The image system and the original system must have the same **boundary condition** on the region of interest. The image system must reproduce the potential of the original system on every boundary of that region where the potential is given, including at infinity. On the surface of a conductor whose total charge is given instead, it must make the surface an equipotential, and the charges that surface encloses, image and real together, must total those it encloses in the original system — the conductor's charge plus any real charge inside it, such as a charge in a cavity. Equal fields on the boundary would also serve, since equal normal components already fix the field. In the above example, the “boundary” is the $z = 0$ plane, and $V(z = 0) = 0$ is satisfied in both systems.
3. More than one image charge may be needed. Because the equation being solved is linear, the fields of the individual images superpose, so images can be added one at a time as further boundaries are brought in, and an image may carry either sign. Exercise 4 below is of this kind.

We shall look at a slightly more complex example now:

Example 5.4: a point charge Q located outside a grounded spherical conducting shell of radius R .

As shown for a cavity in Sec. 5.1, the electric field inside the sphere must be 0, or $\mathbf{E}(r < R) = 0$. For outside the sphere, we take the center of the sphere to be at the origin, Q is located at $(d, 0, 0)$, with $d > R$. We take the image charge $-q$ located at some $(d', 0, 0)$. Thus, the equation for the isosurface $V = 0$ is:

$$0 = \frac{Q}{\sqrt{(x-d)^2 + y^2 + z^2}} + \frac{-q}{\sqrt{(x-d')^2 + y^2 + z^2}}$$

This is equivalent to:

$$q^2 [(x-d)^2 + y^2 + z^2] = Q^2 [(x-d')^2 + y^2 + z^2]$$

or

$$(Q^2 - q^2)(x^2 + y^2 + z^2) + 2(q^2 d - Q^2 d')x = (q^2 d'^2 - Q^2 d'^2)$$

In order for this equation to coincide with the given sphere, we must have:

$$d' = \frac{q^2}{Q^2} d$$

and

$$R^2(Q^2 - q^2) = q^2 d^2 - Q^2 d'^2$$

or

$$q = \frac{R}{d} Q$$

The image is therefore a charge $-q = -RQ/d$ at a distance $d' = R^2/d$ from the center, which lies inside the sphere since $d' < R$ when $d > R$. (The two conditions also admit the solution $q = Q$, $d' = d$, which puts the image on top of the real charge and makes V vanish everywhere, not only on the sphere; rule 1 below excludes it.) Figure 5.6 shows the construction.

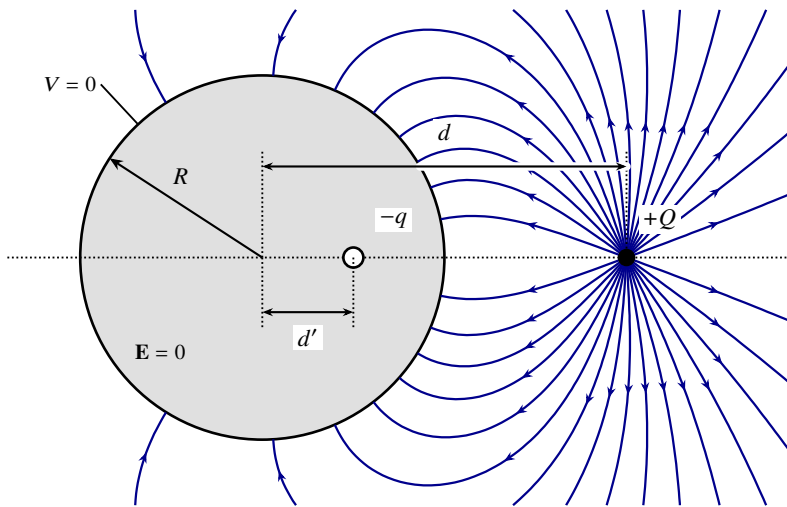


Figure 5.6: Method of image charges: a point charge $+Q$ at distance d outside a grounded spherical conducting shell of radius R . The image charge $-q$ is placed at $d' = R^2/d$ from the center.

5.3 Exercises

1. A conducting spherical ball of radius R_1 is placed concentrically inside a conducting spherical shell with inner radius R_2 and outer radius R_3 ($R_1 < R_2 < R_3$). The inner ball carries charge Q_1 and the outer shell carries total charge Q_2 . Find the electric field in all regions ($r < R_1$, $R_1 < r < R_2$, $R_2 < r < R_3$, $r > R_3$) and the surface charge density on each surface.
2. Two isolated parallel metal plates of the same area carry charges Q_1 and Q_2 . There are no other charges and no external field. Assume that the plates are large enough that edge effects can be neglected.
 - (a) Prove that the charge surface densities on the two facing surfaces are equal in magnitude but opposite in sign, and the charge surface densities on the two outward-facing surfaces are equal in magnitude and same in sign.
 - (b) Calculate the total charge of each surface.
3. Consider again the example of a point charge Q located outside a thin spherical conducting shell of radius R . The distance from the point charge to the center of the shell is d . This time, the spherical shell is not grounded, and instead it has a total charge of Q_0 . Calculate the force exerted on the point charge by the shell.
4. A point charge Q is located at $(a, a, 0)$, where $a > 0$. A grounded conductor fills the whole region where $x < 0$ or $y < 0$ (or both), so that the charge lies in the empty region $x > 0$, $y > 0$. Calculate the force exerted on the point charge by the conductor. (*Hint:* the two conducting walls meet along the z axis, and more than one image charge is needed — see rule 3 above.)
5. A neutral conducting spherical shell, centered at the origin, has inner radius a and outer radius b , where $0 < a < b$. A point charge $q > 0$ is fixed inside the cavity at a distance s from the center, where $0 < s < a$. There are no other charges. Let r be the distance from the center and $\hat{r}$ the unit vector pointing radially outward.
 - (a) Find the total charge on the inner surface and on the outer surface of the shell. Is either of these charges spread uniformly over its surface? Find the electric field for $r > b$.
 - (b) The shell is now connected to the ground, and there are still no charges outside it. Find the charge on the outer surface and the electric field for $r > b$. Does the electric field inside the cavity change?
6. A conducting ball of radius a is centered at the origin and connected to the ground by a thin wire, so that its potential is zero, the same as at infinity. It is surrounded by a thin nonconducting spherical shell of radius $b > a$, also centered at the origin, which carries a fixed total charge $Q > 0$ spread uniformly over its surface. There are no other charges, and the effect of the wire on the field is neglected. Let r be the distance from the origin and $\hat{r}$ the unit vector pointing radially outward.
 - (a) Find the total charge q_g that flows onto the ball from the ground, and the surface charge density σ_g on the ball.
 - (b) Find the electric field $\mathbf{E}$ in the three regions $r < a$, $a < r < b$ and $r > b$.

7. Just outside a conductor in equilibrium the electric field is $(\sigma/\epsilon_0)\hat{n}$, where σ is the local surface charge density and $\hat{n}$ the outward unit normal; inside the conductor it is zero.
- Treat a small patch of the surface, of area ΔA , as a flat, uniformly charged sheet. Using the field of a charged sheet from Chapter 2, find the field that all the other charges produce at the patch, and show that the electric force on the patch's charge is $\sigma^2\Delta A/(2\epsilon_0)$, directed along $\hat{n}$ whatever the sign of σ .
 - An isolated conducting sphere of radius R , centered at the origin, carries a total charge Q . Find the electric force on the charge of its upper half ($z > 0$), and hence the force that the charge on the lower half exerts on the charge on the upper half.
8. Two conducting spheres of radii R_1 and R_2 , where $R_1 > R_2$, are far apart compared with both radii and are joined by a long thin conducting wire. Together they carry a total charge $Q > 0$. Take $V = 0$ at infinity, neglect the charge on the wire, and neglect the effect of each sphere on the other, so that the charge on each sphere is spread uniformly.
- Find the charges Q_1 and Q_2 on the two spheres.
 - Find the ratio σ_1/σ_2 of their surface charge densities and the ratio E_1/E_2 of the field strengths just outside their surfaces. On which sphere is the field at the surface stronger? What does this suggest about the field near a sharp point of a conductor?
9. A point charge $q > 0$ is held at $(0, 0, d)$, a height $d > 0$ above an infinite grounded conducting plane $z = 0$. Use the image-charge solution of Sec. 5.2 for this configuration.
- Find the induced surface charge density $\sigma(x, y)$ on the plane. Show that the total induced charge is $-q$, and find the radius r_0 of the circle $x^2 + y^2 = r_0^2$ that contains half of it.
 - Find the work that an external agent must do to move the charge slowly along the z axis from $z = d$ to infinity. Compare it with the work needed to separate two point charges q and $-q$, initially a distance $2d$ apart, to infinite separation, and explain why the two differ.